

## Project Executable Files

|                      |                                                             |
|----------------------|-------------------------------------------------------------|
| <b>Date</b>          | 15 March 2026                                               |
| <b>Team ID</b>       | SWTID-2026-7169                                             |
| <b>Project Name</b>  | OptiCrop: Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 3 Marks                                                     |

### Project Executable Files

This section requires submission of all executable and deployable files for your project. Ensure all files are properly organized, named, and uploaded to the specified submission platform. The evaluator must be able to run or deploy your solution using the provided files and instructions.

#### Step 1: Submission Checklist

| S.No | Item to Submit                                                  | Submitted (Yes / No / NA)                                           |
|------|-----------------------------------------------------------------|---------------------------------------------------------------------|
| 1    | Complete source code (all files and folders)                    | Yes                                                                 |
| 2    | README / Setup Guide (instructions to run the project)          | Yes                                                                 |
| 3    | requirements.txt / package.json / dependency file               | Yes                                                                 |
| 4    | Database schema / seed files (if applicable)                    | NA — no database used; CSV dataset + model.pkl file storage instead |
| 5    | Environment configuration file (.env.example or similar)        | NA — no environment variables/secrets required                      |
| 6    | Deployed application URL (if hosted)                            | No — application currently runs locally only                        |
| 7    | APK / Executable binary (if applicable for mobile/desktop apps) | NA — web application, not mobile/desktop                            |

|    |                                                       |                                           |
|----|-------------------------------------------------------|-------------------------------------------|
| 8  | Docker file / Containerization config (if applicable) | No — not containerized in current version |
| 9  | Test files and test results                           | Yes                                       |
| 10 | Demo video or walkthrough (if required)               | Yes                                       |

## Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

## Step 3: Deployment / Access Details

| Field                       | Details                                                                                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hosted / Deployed URL       | <a href="https://opticrop-smart-agricultural-production-q73f.onrender.com/">https://opticrop-smart-agricultural-production-q73f.onrender.com/</a>                                                     |
| Login Credentials (Demo)    | [Provide demo username and password for evaluator access]                                                                                                                                             |
| Platform / Hosting Provider | [e.g., Render, Vercel, AWS, Heroku, Azure]                                                                                                                                                            |
| Repository Link             | <a href="https://github.com/Nazneen-1/OptiCrop-Smart-Agricultural-Production-Optimization-Engine.git">https://github.com/Nazneen-1/OptiCrop-Smart-Agricultural-Production-Optimization-Engine.git</a> |
| Demo Video Link             |                                                                                                                                                                                                       |

## Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

**Pre-requisites:** Python 3.x, pip, Anaconda Navigator (recommended) or virtualenv

1. Clone the repository: `git clone <repository-url>`
2. Navigate to the project folder: `cd OptiCrop`
3. Install dependencies: `pip install -r requirements.txt`
4. Ensure model.pkl is present in the /model folder
5. Start the application: `python app.py`
6. Open browser at `http://127.0.0.1:5000/`

## Step 5: Known Issues / Limitations

| S.No | Known Issue / Limitation                                                             | Workaround / Status                                                                        |
|------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 1    | Application currently runs only on Flask's development server (not production-ready) | Plan to deploy using Gunicorn + Nginx or a cloud platform (Heroku/AWS) in future iteration |

|   |                                                                                   |                                                                               |
|---|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| 2 | No user authentication or saved prediction history                                | Out of current project scope; can be added in future versions                 |
| 3 | Model trained on a fixed dataset; does not auto-update with new agricultural data | Future enhancement: enable periodic retraining pipeline with updated datasets |